ITEC 235-940 System Administration

Group Policy Lab

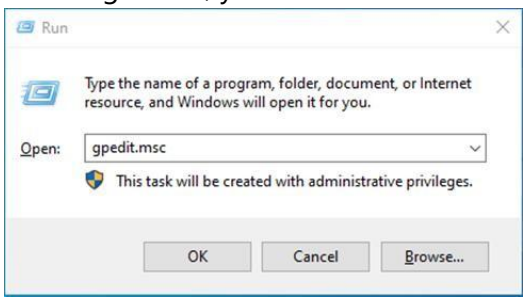
Introduction

In this lab, you will complete several tasks that are commonly performed by a Windows Server administrator. To complete this lab, will need access to both a Windows Server VM and Windows 10/11 VM. The Windows Server VM should be configured as a domain controller and the Windows 10/11 VM should be configured as a domain member. If necessary, you can review previously assigned labs to create the required lab environment.

Exploring Local Group Policy

We will begin with the Local Group Policy that is included with every currently supported version of Microsoft Windows (*except Home Edition*) operating system.

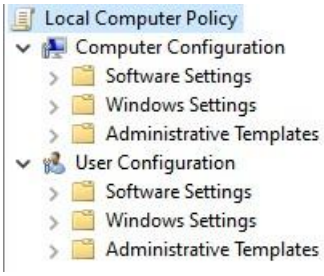
1. Start the Windows 10/11 VM and logon with an account with administrative privileges. Note: If you are logging on with a domain account, it will be necessary for you to start your domain controller (server) as well.
2. Open a Run Command box by holding down the Windows key and pressing R. Once the Run Command box opens, enter `gpedit.msc` and select the "OK" button. Note: If you are using a Mac, you can use the Command key in lieu of the Windows key.

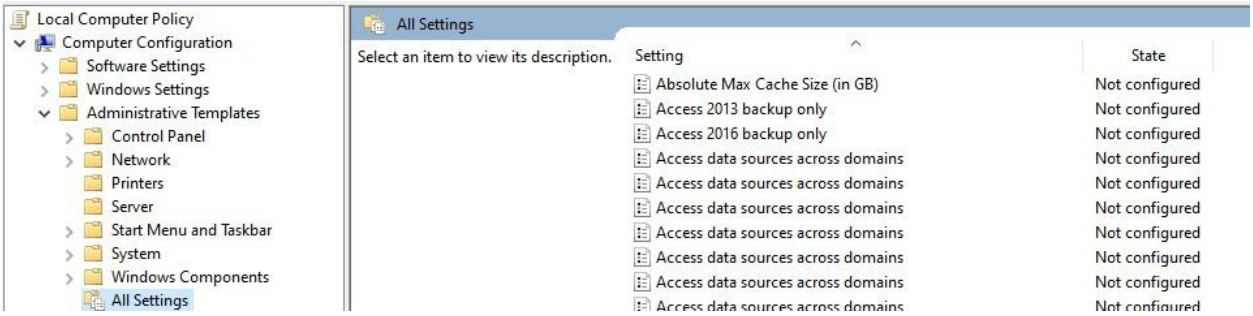


3. When the Local Policy Editor opens, it may be necessary to adjust the vertical bar that separates the panes. This can be accomplished by hovering your mouse pointer over the vertical separator until the mouse pointer appears as a double arrow. Once the double arrow appears, hold down the left mouse button and drag the vertical separator to the left or right as needed.

4. Notice that Group Policy settings are separated into two categories labeled Computer Configuration and User Configuration. Notice that both categories contain the same settings. This lab focuses on the settings contained in the Computer Configuration category.

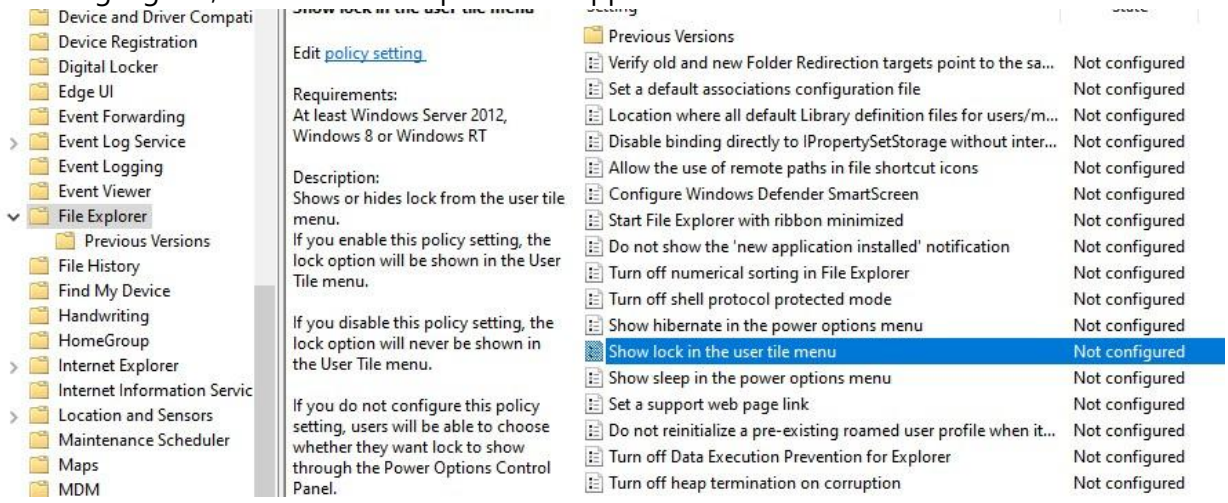
5. Under “Computer Configuration”, click on the arrow to the left of “Administrative Templates” to expand the listing. Click on “All Settings” in the left pane to view the list of available settings in the right pane. Take a moment to scroll through the list of options.



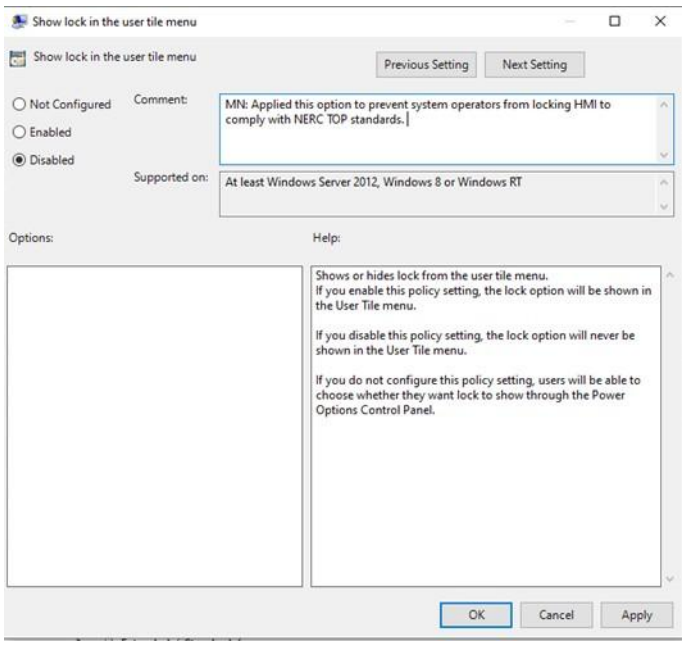


6. Under “Computer Configuration”, expand “Administrative Templates”, “Windows Components”, and “File Explorer”.

7. In the pane on the right, click “Show lock in user tile menu”. If you click on it one time to highlight it, notice the description that appears on the left.

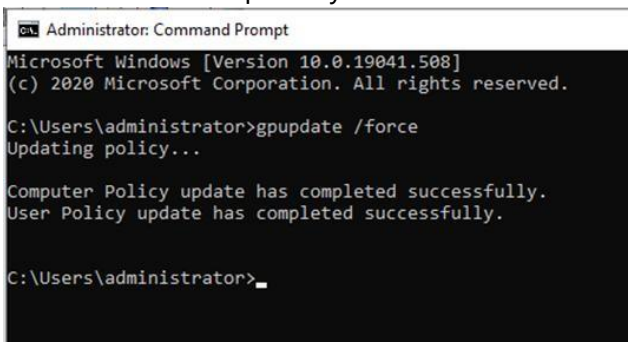


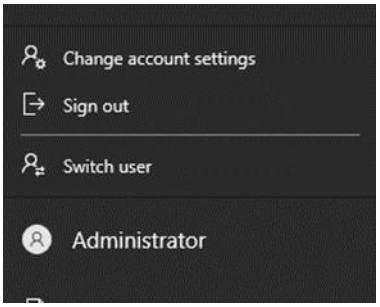
8. Double click “Show lock in the user tile menu” to open the policy. Read the “Help:” section in the lower right corner for a description of the policy setting options. Notice the “Comment:” field. The Comment field can be used to document notes regarding the policy configuration. Click the “Disabled” radio button to remove the Lock option from the Start Menu. This will prevent our users from locking their computers. Click “Apply” to apply the setting.



9. Click on the Windows Start button and then click on the logged-on user, notice that "Lock" is still an option. Recall from the reading that Group Policy changes are automatically updated at a pre-configured interval (the default interval is 90 minutes). Instead of waiting, you can manually update Group Policy using the gpupdate command. Open a command window and enter the command `gpupdate /force` and press the Enter key. Once complete, verify that the "Lock" option has been removed.

   In the next section, we will configure the same policy on the domain controller. However, instead of disabling the Lock feature we will enable it to demonstrate how Group Policy overrides Local Group Policy.

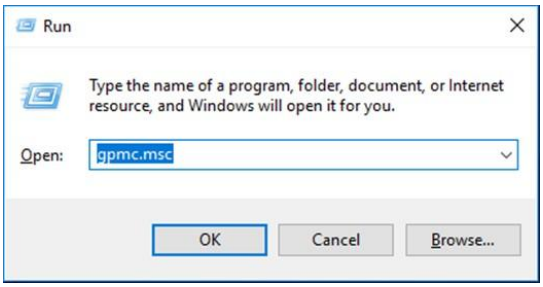




Exploring Domain Group Policy

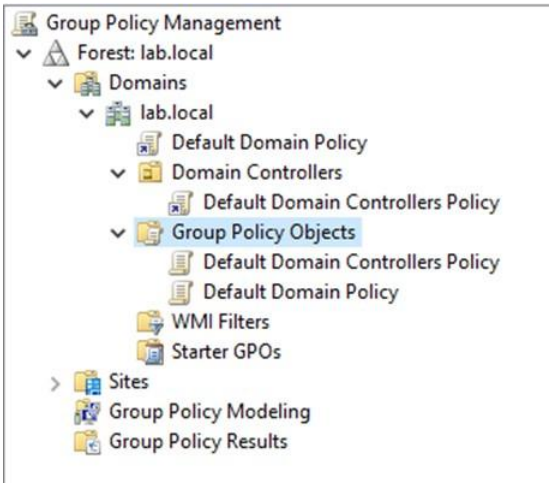
In the previous section, you used the Local Group Policy on the Windows 10/11 desktop computer to remove the "Lock" option. In this lab, you will configure the same policy at the domain level to enable the "Lock" option to demonstrate the hierarchy of Group Policy in a

domain setting. We will also look at a few security settings and finish the lab by backing up our settings.

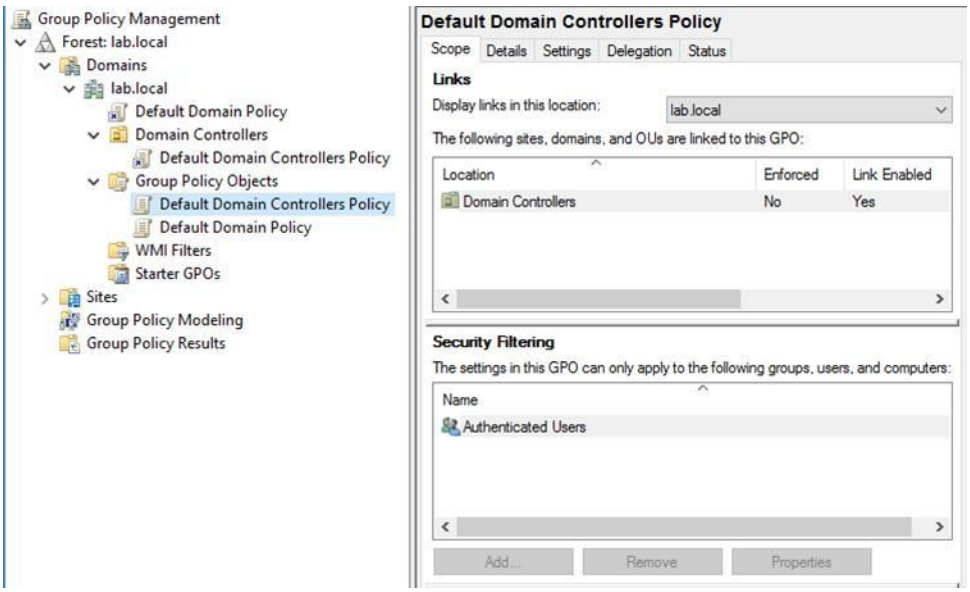


1.Start your Server VM, and logon with an account configured with domain admin privileges.

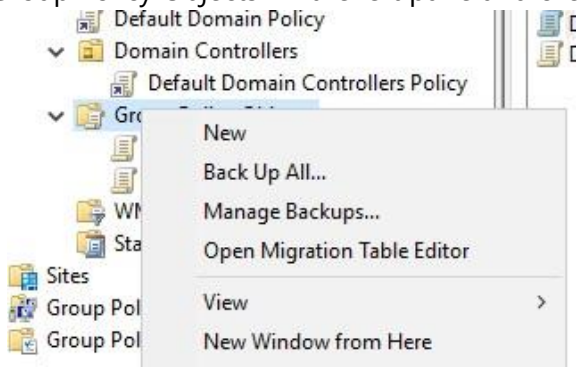
2.Open a Run Command box by holding down the Windows key and pressing R. Once the Run Command box opens, type `gpmc.msc` and press the Enter key. This will start the Group Policy Management Console. Note: If you are using a Mac, you can use the Command key in lieu of the Windows key.



3. Click on the arrow to the left of the Forest to expand it.
4. Click on the arrow to the left of Domains to expand it.
5. Click on the arrow to the left of your unique domain name to expand it.
6. Notice the two default GPOs titled "Default Domain Controllers Policy" and "Default Domain Policy." These two GPOs are stored in the "Group Policy Objects" Organizational Unit (OU). The Default Domain Policy is linked to the root of the domain and the Default Domain Controllers Policy is linked to the Domain Controllers OU.
7. Click on "Default Domain Controllers Policy" in the "Group Policy Objects" OU. Click on the "Scope" tab in the pane on the right. You will notice that this policy applies to the "Domain Controllers" Active Directory OU and "Authenticated Users" Active Directory OU.

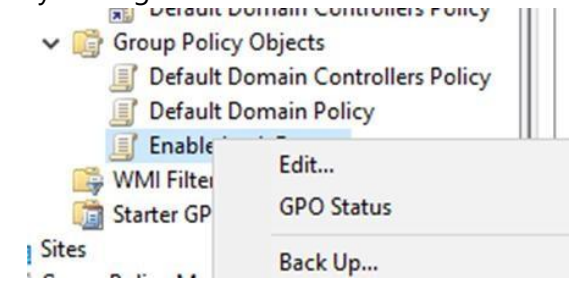


8. To create a new GPO and apply it to the Windows 10/11 workstation, right click on "Group Policy Objects" in the left pane and click on "New".



9. In the "Name:" field, enter "Enable Lock Button" and click the "OK" button.

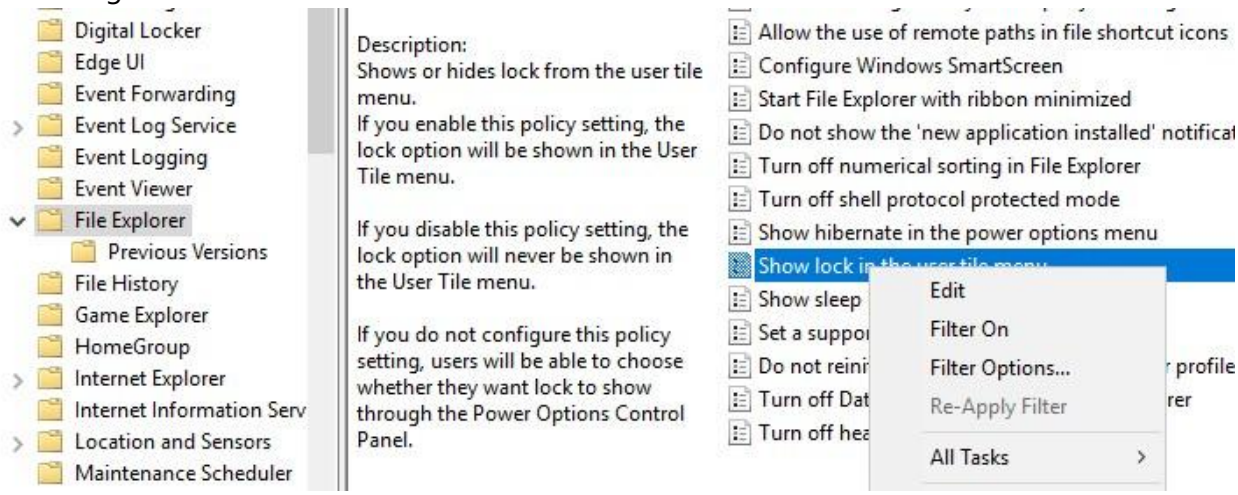
10. Right click on the "Enable Lock Button" GPO and left click on "Edit" to open the Group Policy Management Editor.



11. When the Group Policy Management Editor opens, you will notice the same two categories of Group Policy settings, Computer Configuration and User Configuration, that were in the Local Group Policy Settings.

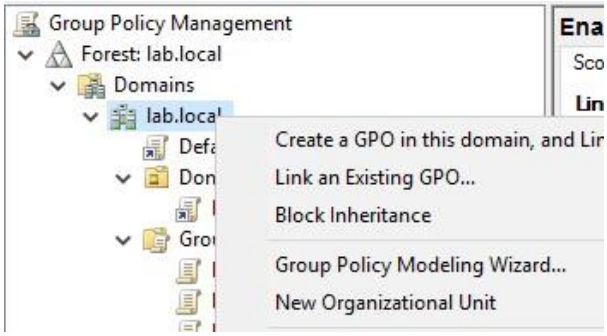
12. Under Computer Configuration, expand Policies, Administrative Templates, and Windows Components. Click on “File Explorer”.

13. Right click on “Show lock in user tile menu” and then left click on “Edit”.

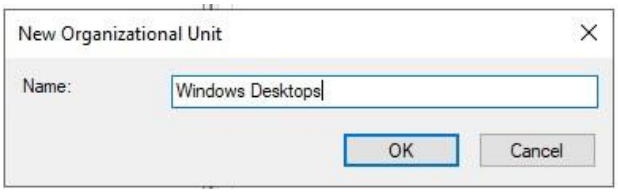


14. Select the radio button for “Enabled” to enable the Lock option on the Start Menu. Click the “OK” button and close the Group Policy Management Editor window.

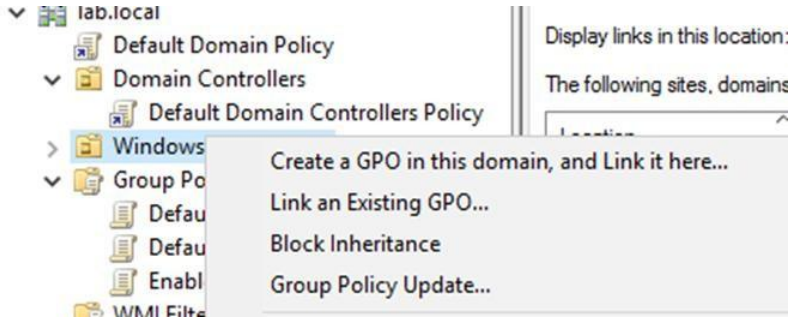
15. Right click on your unique domain name and left click “New Organizational Unit”.



16. Enter “Windows Desktops” in the “Name:” field and click the “OK” button.



17. Right click on the newly created OU named Windows Desktops and left click “Link an Existing GPO.”

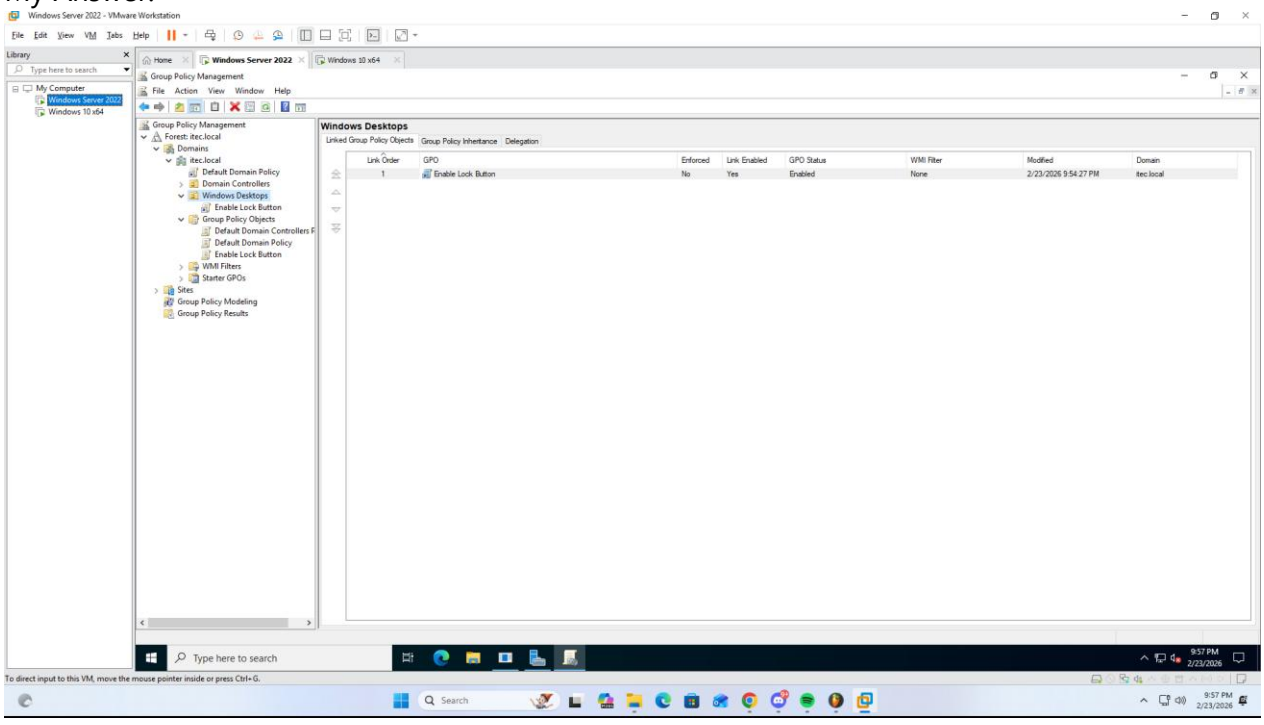


18. Select “Enable Lock Button” from the “Group Policy Objects:” pane and click the “OK” button.

19. Expand the “Windows Desktops” OU to view the linked “Enable Lock Button” GPO. Create a screen capture to submit for completion of the lab.



My Answer:

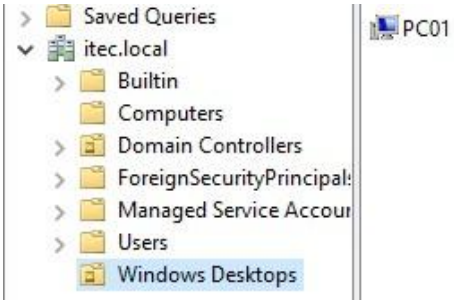


20. Click the Windows Start button and then scroll down until you see "Windows Administrative Tools".

21. Click on "Windows Administrative Tools" and then scroll down until you see "Active Directory Users and Computers".

22. Click on "Active Directory Users and Computers".

23. Notice that a new OU named "Windows Desktops" appears within Active Directory. Click on the "Computers" OU and drag the Windows 10/11 workstation to the "Windows Desktops" OU. Accept the warning. Close Active Directory.



24. Logon to the Windows 10/11 VM and perform a manual Group Policy update (Hint: gpupdate /force). Once complete, click the start button.

25. Click on your user account image. Notice that the "Lock" option has returned. Why did this happen? Remember that GPOs linked to the Domain OU take precedence over Local GPOs.

Creating Backups of GPOs

It is common for Group Policy Settings to accumulate overtime. As a Windows Server Administrator, it is your responsibility to keep these settings organized and backed-up.

1. Open the Windows Server VM.

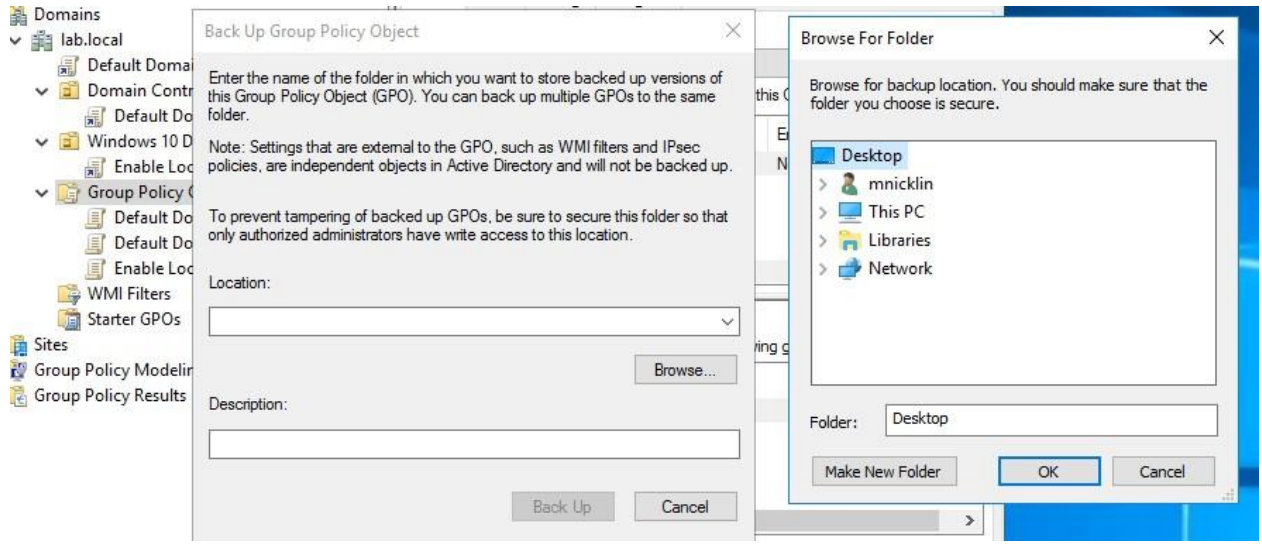
2. Open the Group Policy Management Console (Hint: gpmc.msc).

3. Right click on the "Group Policy Objects" OU and then left click on "Back Up All…".

4. Click the "Browse" button and select a location for the backup.

5. Click the "OK" button.

That's it! Alternatively, you can right click on a single GPO and select "Back Up….".



This lab provided you with a brief overview of Group Policy, but an entire semester devoted to Group Policy is necessary to cover it in-depth. You are encouraged to continue experimenting with the many options available to you within Group Policy. For example, Group Policy can be used to deploy software (*.msi only*), run scripts, configure Windows Firewall, and change desktop wallpaper.